

PHASE 5

Molecular Quantum Soft Spaces

Preregistered Whitepaper for Benchmark Design, Predictive Utility and
Computational Value

Version 1.0
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**Phase 5 is not designed to demonstrate that Soft Spaces exist.
It is designed to test whether the previously established structure provides predictive
or computational value in a molecular quantum problem.**

Status: PRE-EXPERIMENT / PREREGISTRATION

Planned implementation series: v50_x

1. Executive Summary

Phase 5 moves the Soft Spaces program from structural and hardware-proxy validation into a domain-specific molecular quantum benchmark. The scientific question is deliberately stricter than in earlier phases: can information associated with Soft-Space structure improve the selection, ranking, or prioritization of molecular quantum states or candidate configurations under a fixed computational budget?

The primary benchmark is a linear H4 chain in a minimal STO-3G basis, represented in an eight-spin-orbital active space. H4 is selected because it is small enough for essentially exact classical reference calculations while still exhibiting geometry-dependent correlation and near-degeneracy. This creates a controlled setting in which a Soft-Space signal can be tested against nontrivial molecular structure rather than against a toy two-qubit example.

The first Phase 5 claim will therefore be narrow: Soft-Spaces guidance is useful only if it improves a prespecified task relative to prespecified baselines, survives matched NULL controls, and remains useful across more than one molecular geometry. No claim of quantum advantage, industrial advantage, or chemical discovery is permitted from the first benchmark alone.

Primary decision question: Under the same evaluation budget, does Soft-Spaces-guided selection recover more chemically relevant low-energy structure than baseline selection?

2. Scientific Transition from Phase 4 to Phase 5

Earlier phase	Primary question	Phase 5 consequence
Phase 3	Is there a reproducible REAL-vs-NULL structure and a stable hotspot concept?	Treat the structural phenomenon as a prior hypothesis, not as the endpoint.
Phase 4	Does the signal retain utility under application screens and noisy/hardware-proxy conditions?	Require budget-matched application metrics, not only structural metrics.
Phase 5	Does the structure predict or prioritize useful molecular quantum information?	Success requires molecular task value relative to classical and unguided baselines.

The key methodological constraint is separation between discovery and confirmation. Phase 5 should not tune its PASS criteria after observing H4 outcomes. The preregistration and the first reference data products must therefore be generated before optimization of any Soft-Spaces-specific scoring rule.

3. Scope and Non-Claims

Phase 5 v1.0 is scoped to a computational benchmark, not to a claim of practical quantum advantage. Specifically:

- The study tests predictive or computational value, not merely existence of hotspots.
- The primary comparison is budget-matched selection/ranking performance.
- Exact or near-exact classical results are used as reference truth wherever tractable.
- The study does not claim superiority over mature quantum-chemistry packages in production workloads.
- The study does not claim fault-tolerant resource savings unless a later resource analysis explicitly demonstrates them.
- The study does not infer industrial value from a single molecule or a single geometry.

4. Primary Molecular Benchmark

4.1 System: linear H4 chain

The first benchmark will use four hydrogen atoms arranged linearly with equal nearest-neighbour spacing R . In a minimal STO-3G basis, the natural starting representation contains four spatial orbitals and eight spin orbitals, corresponding to an eight-qubit occupation-number encoding before optional symmetry reductions.

The central advantage of H4 is controllable difficulty. By scanning R , the benchmark can move from a more weakly correlated regime toward stretched geometries with stronger static correlation. The molecular Hamiltonian therefore changes continuously while the Hilbert-space dimension remains small enough for exact diagonalization and exhaustive validation.

Frozen item	Phase 5 v1.0 choice
Molecule	Linear H4
Basis	STO-3G
Electrons	4
Spin orbitals / nominal qubits	8
Geometry family	Equal-spacing linear chain, spacing R
Reference solver	Full configuration interaction / exact diagonalization in the chosen active space
Initial geometry set	At least 5 prespecified R values spanning weak-to-stronger correlation
Primary representation	Occupation-number computational basis; fermion-to-qubit mapping frozen before production runs

Recommended frozen geometry grid: $R = 0.8, 1.0, 1.4, 1.8, \text{ and } 2.4 \text{ \AA}$. These values may be adjusted once, before any Soft-Spaces-vs-baseline comparison, if numerical conditioning or orbital-definition issues require it. Any adjustment must be logged as a preregistration amendment.

4.2 Secondary benchmark after primary lock

LiH in STO-3G with a prespecified frozen-core/active-space reduction is reserved as a secondary validation problem. It must not be used to tune the primary H4 decision thresholds. Its purpose is transfer: determine whether any H4 result survives a chemically different Hamiltonian and orbital structure.

5. The Concrete Benchmark Task

The first benchmark is a state/configuration prioritization task. For each geometry, the exact molecular ground state defines a reference distribution over computational-basis determinants. Candidate determinants are then ranked by a Soft-Spaces-derived score and by several baselines. Each method receives the same top-K or evaluation budget.

Target quantity: recover and prioritize the determinants that carry the largest reference-state weight and/or the greatest contribution to a prespecified low-energy reconstruction objective.

Two task views are allowed, but the primary one must be frozen before production:

1. Support recovery: identify high-weight determinants of the exact ground-state wavefunction.
2. Low-energy reconstruction: use the selected determinant subset to form a truncated subspace and measure the energy error relative to the exact active-space ground-state energy.

The low-energy reconstruction view is scientifically stronger because it connects ranking quality to a molecular observable. Support recovery remains useful as a diagnostic because it shows whether the method is finding relevant basis structure rather than benefiting from a single numerical coincidence.

6. Preregistered Hypotheses

ID	Hypothesis
H1 - Predictive enrichment	At fixed K, Soft-Spaces-guided ranking enriches chemically relevant determinants relative to random and simple heuristic baselines.
H2 - Energy utility	A truncated variational subspace selected by Soft-Spaces guidance reaches lower ground-state energy error than budget-matched unguided selection.
H3 - REAL vs NULL specificity	The gain is larger for the frozen REAL construction than for matched NULL controls generated under the same pipeline.
H4 - Geometry robustness	The effect is not confined to one H4 bond length; it persists across a prespecified fraction of the geometry grid.
H5 - Transfer	If H1-H4 pass, the direction of effect transfers to the secondary molecular benchmark without retuning the core scoring rule.

7. What Soft Spaces Must Predict

Phase 5 deliberately avoids a vague claim that “hotspots matter.” A usable Soft-Spaces signal must map to a measurable molecular prediction. For each candidate determinant or local region of the encoded state space, the pipeline must produce a frozen score before accessing the target reference label used for evaluation.

- Primary prediction: relative priority of candidate determinants for inclusion in a restricted low-energy subspace.
- Secondary prediction: probability or rank of high-amplitude determinants in the exact ground state.
- Exploratory prediction: sensitivity of a candidate region to controlled Hamiltonian perturbations or parameter changes.
- Forbidden leakage: exact ground-state amplitudes, exact energy contributions, or any target-derived quantity may not be used to define the predictor being evaluated.

8. Prespecified Baselines

A positive result is meaningful only against explicit comparators. The initial minimum baseline suite is:

Baseline	Purpose
Random-K	Chance-level selector; repeated with many seeds to establish a distribution.
Uniform / lexicographic control	Deterministic non-informative ordering check.
Hartree-Fock proximity	Prioritize the Hartree-Fock determinant and low excitation-rank configurations.
Diagonal-energy heuristic	Rank determinants using the diagonal matrix element $\langle D H D \rangle$ only.
Perturbative / coupling heuristic	A simple frozen score based on coupling to the Hartree-Fock reference or a first-order/second-order motivated proxy, if well-defined for the active space.
NULL Soft Spaces	Matched control using the Phase 5 NULL-generation rule, identical downstream evaluation.

The exact baseline implementation must be frozen in code and written to a machine-readable preregistration artifact. If an additional strong baseline is introduced later, it may strengthen the study but may not replace a failed preregistered comparison post hoc.

9. Metrics and Budget Matching

Metric	Definition / role
Precision@K / Recall@K	Overlap between selected candidates and a prespecified set of high-weight reference determinants.
Captured probability mass	Sum of exact ground-state determinant probabilities contained in the selected set.
Subspace energy error	$ E_K - E_{\text{exact}} $ after diagonalization in the selected determinant subspace.
NDCG@K or rank correlation	Ranking quality when the full target ordering is informative.
Evaluation cost	Number of candidates, Hamiltonian matrix elements, circuit evaluations, shots, or equivalent task-relevant calls consumed by the method.
REAL-NULL delta	Difference in the primary metric between REAL and matched NULL pipelines under identical budget.

All headline comparisons are budget matched. If Soft-Spaces construction itself consumes a non-negligible computational resource, that cost must be reported separately and, in later industrial analyses, included in total cost-of-solution accounting.

10. PASS / FAIL Gates

The following gates are intentionally conservative. Phase 5 v1.0 is a benchmark-validation study; a single favorable plot is insufficient. The exact numerical thresholds should be encoded in `v50_0_preregistration.py` before the first production comparison.

Gate	PASS criterion	FAIL interpretation
P1 - Primary utility	Soft-Spaces beats Random-K on the primary metric at ≥ 4 of 5 frozen H4 geometries, with a positive aggregate effect.	No robust evidence of useful enrichment.
P2 - Nontrivial baseline	Soft-Spaces exceeds at least one chemically informed non-random baseline on the primary metric in ≥ 3 of 5 geometries.	Signal may be no better than simple chemistry heuristics.
P3 - REAL vs NULL	Aggregate REAL performance exceeds matched NULL, with directionally positive delta in ≥ 4 of 5 geometries.	Observed utility is not specific to the claimed structure.
P4 - Energy relevance	For a prespecified K or budget range, selected-subspace energy error is lower than Random-K median in ≥ 4	Ranking advantage does not translate to the molecular objective.

	of 5 geometries.	
P5 - Robustness	No single geometry contributes > 60% of the aggregate improvement, and the sign of the effect is stable across seeds/controls.	Result is dominated by one special case.
P6 - Transfer gate	Only attempted after P1-P5. Direction of effect is retained on the secondary benchmark without retuning the core score.	Primary result does not yet generalize.

Overall Phase 5 v1.0 status is PASS only if P1, P3 and P4 pass. P2 and P5 are required for a stronger “molecular utility” claim. P6 is a separate transfer milestone.

11. Statistical and Reproducibility Controls

- Use a fixed master seed list for all stochastic baselines and NULL constructions.
- Report per-geometry values, not only averages.
- Use bootstrap confidence intervals or paired resampling across matched seeds where appropriate.
- Predefine all K values or budget points used in headline claims.
- Separate confirmatory metrics from exploratory plots.
- Store Hamiltonian fingerprints, geometry, basis, mapping, active-space choices and software versions with every result.
- Write immutable CSV/JSON result tables before generating summary figures.
- Record any preregistration amendment before reading the affected benchmark result.

12. What Counts as Industrial Value

Industrial value is not established by statistical significance alone. A later application claim must connect the signal to a cost or capability relevant to molecular simulation workflows. Phase 5 therefore defines industrial value operationally rather than rhetorically.

Value category	Evidence required
Search-space reduction	Fewer determinants/configurations evaluated for the same target accuracy.
Circuit reduction	Fewer state preparations, ansatz terms, measurements, or circuit evaluations for the same accuracy.
Shot reduction	Lower total shot count at fixed uncertainty/decision quality.
Classical preselection	Reduced classical enumeration or matrix construction cost without unacceptable loss of accuracy.
Robustness	Advantage survives moderate noise/model misspecification

	and is not dependent on one fine-tuned geometry.
Transferability	Benefit appears in more than one molecular family or task without re-engineering the rule.

A credible industrial statement therefore requires a cost-normalized comparison against strong alternatives. The first H4 experiment can establish proof of utility, not industrial readiness.

13. Phase 5 Execution Plan

Version	Purpose	Output
v50_0_preregistration.py	Freeze benchmark constants, geometry grid, seeds, metrics, PASS/FAIL gates and file schema.	Machine-readable preregistration JSON + human-readable manifest.
v50_1_reference_h4.py	Generate H4 Hamiltonians and exact/FCI reference states for all frozen geometries.	Reference energies, determinant weights, Hamiltonian hashes.
v50_2_softspace_mapping.py	Apply the frozen Soft-Spaces construction to the molecular encoding without target leakage.	Per-candidate Soft-Spaces scores + NULL controls.
v50_3_baselines.py	Generate all baseline rankings under the same candidate universe.	Baseline score tables.
v50_4_primary_benchmark.py	Evaluate support recovery and selected-subspace energy performance.	Primary result tables + gate decisions.
v50_5_robustness.py	Seed, geometry, parameter and sensitivity checks.	Robustness report.
v50_6_secondary_transfer.py	Apply frozen pipeline to LiH or another preregistered secondary molecule.	Transfer result; no retuning.

14. Recommended Repository Layout

Phase5/V_1_0/

```
code/          v50_x scripts
data/reference/ molecular Hamiltonians and exact reference data
data/preregistration/ frozen JSON/YAML manifests
results/raw/    immutable raw benchmark tables
results/summary/ aggregate tables and gate decisions
figures/        generated plots only
logs/           execution logs and environment capture
docs/whitepapers/ this document and later revisions
```

15. Minimum Preregistration Record

Before v50_1 is treated as a production run, v50_0 should write a record containing at least:

- Project/version identifiers and Git commit hash.
- Molecule, geometry grid, charge, multiplicity, basis set and units.
- Active-space and orbital-selection rule.
- Fermion-to-qubit mapping and any symmetry/tapering policy.
- Candidate universe and excluded states.
- Soft-Spaces score definition and every tunable parameter.
- NULL-generation rule and seed list.
- Baseline definitions.
- Primary/secondary metrics and K/budget grid.
- PASS/FAIL thresholds.
- Reference solver and numerical tolerances.
- Software environment and dependency versions.
- Output file schema and hash policy.

16. Main Scientific Risks and Mitigations

Risk	Why it matters	Mitigation
H4 is too small	A useful signal may be hidden by near-exhaustive tractability.	Treat H4 as validation; move to a larger active space only after the pipeline is frozen.
H4 is too special	Regular symmetry may exaggerate structure.	Use asymmetric geometry perturbations as a robustness test and a second molecule for transfer.
Target leakage	Using exact amplitudes to shape the predictor invalidates the benchmark.	Separate predictor construction from reference labels; hash/freeze artifacts.
Baseline weakness	Beating random alone is not meaningful.	Require chemically informed baselines and budget matching.
Hyperparameter overfit	Tuning on the same geometries inflates apparent value.	Freeze parameters before primary evaluation; reserve transfer benchmark.
Metric cherry-picking	Different metrics can reverse conclusions.	Name one primary metric and one primary K/budget region before production.
Computation cost hidden	A strong ranking may cost more than it saves.	Track Soft-Spaces construction cost separately and later include total cost.

17. Decision Logic After the First Benchmark

The project should branch according to observed evidence, not according to a desire for a positive result:

3. If P1/P3/P4 pass: proceed to robustness, transfer and stronger cost accounting.
4. If Random-K is beaten but informed baselines are not: classify the result as structural enrichment, not molecular advantage; improve the hypothesis rather than the rhetoric.
5. If REAL does not beat NULL: stop the application claim and diagnose whether the Phase 3/4 signal is mismatched to the molecular encoding.
6. If only one geometry works: treat it as a case study and test whether the effect tracks correlation regime, symmetry, or another physical property.
7. If all primary gates fail: record a clean negative result. The failure itself constrains what Soft Spaces can and cannot be used for.

18. Allowed Claim Language

Evidence level	Permitted wording
Before benchmark	“We test whether Soft-Space structure provides predictive or computational value in a molecular quantum benchmark.”
P1 only	“Soft-Spaces guidance enriches selected molecular basis states relative to random selection in the tested H4 benchmark.”
P1 + P3 + P4	“The preregistered REAL Soft-Spaces pipeline improves budget-matched H4 subspace selection relative to random selection and matched NULL controls.”
P1-P5	“The effect is robust across the preregistered H4 geometry set and exceeds at least one nontrivial chemistry-informed baseline.”
P6 transfer	“The direction of the preregistered effect transfers to a second molecular benchmark without retuning the core scoring rule.”

Terms such as “quantum advantage,” “industrial advantage,” “chemical discovery,” or “fault-tolerant resource advantage” require additional dedicated evidence and are outside the scope of this whitepaper.

19. Phase 5 v1.0 Conclusion

Phase 5 should begin as a disciplined falsification-oriented benchmark. The central object is not a new visualization of Soft Spaces but a measurable decision advantage under fixed resources. Linear H4 provides an appropriate first molecular test because it combines exact classical verifiability, an eight-spin-orbital state space, and a tunable correlation regime.

The next artifact after this whitepaper should therefore be v50_0_preregistration.py. Its purpose is to freeze the geometry grid, representation, candidate universe, seeds, metrics, baselines and decision gates before the reference and Soft-Spaces result files are compared.

Phase 5 succeeds scientifically even if the result is negative, provided the benchmark is frozen in advance and the conclusion follows the prespecified evidence gates.

Appendix A. Frozen Phase 5 v1.0 Decisions

Item	Decision
Phase name	Phase 5 - Molecular Quantum Soft Spaces
Version family	v50_x
Primary molecule	Linear H4
Basis	STO-3G
Nominal encoding size	8 spin orbitals / 8 qubits before optional symmetry reduction
Primary task	Budget-matched determinant/subspace prioritization
Primary evidence	Selection quality + low-energy subspace reconstruction
Required controls	Random, deterministic non-informative, chemistry-informed, and matched NULL
Primary gate logic	P1 + P3 + P4 required for Phase 5 v1.0 PASS
Secondary molecule	LiH active-space benchmark after primary lock
First code artifact	v50_0_preregistration.py